Residual and Jet Certificates for a Local Penrose Inequality

A source-specific Ram Calculus audit of manuscript M517

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Abstract

We analyze the pinned manuscript on conformal perturbations of positive-mass Schwarzschild–anti-de Sitter data, retaining its fixed-seed quantifiers, MOTS boundary convention, Lorentz mass normalization, and exterior decay. The source's quadratic and quartic positivity arguments are separated from their actual-branch remainder requirements. We prove an exact extension of the static TT square to an inexact potential: its overestimate is precisely the sum of a weighted scalar-equation residual square and a boundary inverse-operator residual square. We also give explicit residual-to-branch and branch-to-deficit bounds, retain every third- and fourth-order conformal coefficient, and derive a residual version of the source's dipole obstruction. A conditional rotation-invariant normal form clarifies the quartic coefficient on the exact kernel and the limitations of near-kernel conclusions. The Ram correspondences concern physical Gram metrics, complete differentiated jets, and explicit error accounting, not a replacement of exterior geometric analysis by finite matrices. No numerical seed, speedup, independent proof of the full source theorem, or historical novelty is asserted. Contribution assessment: C3/5, exact source-specific residual identities and explicit branch error bounds give a notable local verification improvement while the global quartic geometry and numerical certification remain unverified.

Contents

1 Pinned problem and exact conventions **3**

2 Stepwise source dependency audit **4**
2.1 Analysis and coefficient-to-branch passage 4
2.2 CMC jets, quadratic cancellation, and exact kernel 4
2.3 Fourth order and radial equality 5

3 Exact inexact-potential TT identity **6**
3.1 Constraint defects and finite Gram matrices 7

4 Explicit residual-to-branch and mass bounds **8**
4.1 The convergent observable and its error 9
4.2 Residual size, explicit radii, and end tails 9

5 Complete coefficient recursion and its checks **10**

6 Residual form of the dipole obstruction **11**

7 Quartic symmetry and the near-kernel boundary **12**

8 Specific private Ram correspondences **13**

9 Public comparison and research boundaries **14**

10 Verification record and contribution assessment **15**

1 Pinned problem and exact conventions

The source is OpenAI, *A local Penrose inequality for conformal perturbations of Schwarzschild–anti-de Sitter data*, dated October 5, 2026, at catalogue commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The local source directory, relative to the study root

`C:/Users/mural/OneDrive/Documents/OpenAI-722-math/`,

is

`_sources/openai-math/preprints/A-local-Penrose-inequality-for-conformal-perturbations-of-Schwarzschild-anti-de-Sitter-data-October-5-2026/`.

Throughout, "source" refers to that version, not a subsequently updated public manuscript. All seven substantive section files have been read in full, including their proofs, finite-jet constructions, and tensor algebra. There is no separate mathematical appendix in the supplied source tree. The source preamble, bibliography, and README have also been inspected. Input hashes and exact private reading scopes accompany this report in the author audit; the catalogue row is a routing and pinning record, not mathematical evidence in place of these readings.

Fix $m_0 > 0$, let $a > 0$ solve $2m_0 = a(1 + a^2)$, and put

$$\begin{gathered} M = [0,\infty) \times S^2, \quad S = \{0\} \times S^2, \quad g_0 = \mathrm{d}s^2 + r^2\sigma, \\ r(0) = a, \quad N = r', \quad N^2 = 1 + r^2 - 2m_0/r. \end{gathered} \tag{1.1}$$

Here $N > 0$ for $s > 0$, $N(0) = 0$, σ has Gauss curvature one, and $n = \partial_s$ points into the exterior. The integration-domain outward normal at S is $-n$. The Laplacian is $\Delta = \operatorname{div} \nabla$, with nonpositive eigenvalues on the sphere. The cosmological constant is -3. Define

$$\kappa = N'(0) = \frac{1 + 3a^2}{2a}, \quad B = \mathrm{Ric}_{g_0} + 3g_0, \quad L = \Delta_{g_0} - 3, \quad D = -\Delta_{a^2\sigma} + 2\kappa/a. \tag{1.2}$$

In a radial orthonormal splitting,

$$B_s = 1 - 2m_0/r^3, \quad B_t = 1 + m_0/r^3, \quad \operatorname{Hess} N = NB, \quad \operatorname{tr} B = 3, \quad \operatorname{div} B = 0, \quad LN = 0. \tag{1.3}$$

The boundary eigenvalues are $D_\ell = \ell(\ell + 1)/a^2 + 2\kappa/a > 0$.

For every fixed $0 < \alpha < 1$, $3/2 < \tau < 3$, the source considers a smooth symmetric seed $q \in C^{1,\alpha}_\tau$ with $\operatorname{tr}_{g_0} q = \operatorname{div}_{g_0} q = 0$. The weighted space means that tensor components, their covariant derivatives through the stated order, and weighted local Hölder seminorms decay as $e^{-\tau s}$ on unit hyperbolic balls; ordinary regularity is used on compact sets. The reference end metric is $b = \mathrm{d}r^2/(1 + r^2) + r^2\sigma$. Tensor norms below use g_0 unless stated otherwise. The source assumes a positive solution branch with

$$\Delta\phi_\epsilon = \tfrac{3}{4}(\phi_\epsilon^5 - \phi_\epsilon) - \tfrac{\epsilon^2}{8}|q|^2\phi_\epsilon^{-7}, \tag{1.4}$$

$$\partial_s\phi_\epsilon|_S = \tfrac{\epsilon}{4}q_{ss}\phi_\epsilon^{-3}, \quad \phi_0 = 1, \quad \|\phi_\epsilon - 1\|_{C^{2,\alpha}_\tau} \longrightarrow 0. \tag{1.5}$$

The seed is independent of ϵ. Existence for every seed is not a conclusion of the source theorem. No differentiability in a stronger weighted space is assumed. The actual data are $g_\epsilon = \phi_\epsilon^4 g_0$ and $K_\epsilon = \epsilon\phi_\epsilon^{-2}q$. The convention $\partial_t g = 2K$ in Gaussian time gives $\theta_+ = H + \operatorname{tr}_S K = 0$ at S, since $H = \epsilon\phi_\epsilon^{-6}q_{ss} = -\operatorname{tr}_S K$.

Set $V_0 = \sqrt{1+r^2}$, $V_i = rx_i$ and $e = g_\epsilon - b$. The mass is fixed by

$$p_\mu = \frac{1}{16\pi} \lim_{R\to\infty} \int_{r=R} \left[V_\mu(\mathrm{div}_b\, e - \mathrm{d}\,\mathrm{tr}_b\, e) + (\mathrm{tr}_b\, e)\mathrm{d}V_\mu - e(\nabla^b V_\mu, \cdot) \right] (\nu_b)\mathrm{d}A_b. \tag{1.6}$$

Finiteness and future timelikeness for small positive ϵ are hypotheses. The spatial components are not angular momentum. Write

$$\begin{gathered} m_{\mathrm{AH}} = \sqrt{p_0^2 - |p_{\mathrm{sp}}|^2}, \quad b_*(A) = \sqrt{A/(16\pi)}(1 + A/(4\pi)), \\ \mathfrak{D}_q = m_{\mathrm{AH}} - b_*(A_\epsilon), \quad A_\epsilon = \int_S \phi_\epsilon^4 \mathrm{d}A_0. \end{gathered} \tag{1.7}$$

Thus $p(0) = (m_0, 0, 0, 0)$, $A_0 = 4\pi a^2$ and $b_*'(A_0) = \kappa/(8\pi)$. The exact source claim is: *for each* background, fixed seed and admitted branch, *there exists* $\epsilon_0 > 0$ such that $\mathfrak{D}_q(\epsilon) \geq 0$ for $0 \leq \epsilon < \epsilon_0$, with equality for radial seeds and strict inequality otherwise at sufficiently small positive parameters. No uniform ϵ_0 over seeds is asserted. The primary theorem uses the area of S itself, without outermostness, outer area-minimization, or a sign on q_{ss}. The source's additional enclosing-area corollary imposes all of those appropriate extra conditions, including exclusion of weakly outer trapped enclosing surfaces; this report does not supply them.

2 Stepwise source dependency audit

We distinguish a checked local derivation from a read dependency. Reproducing an equation, reading its proof, or passing an algebra test is not independent verification of its entire geometric use.

2.1 Analysis and coefficient-to-branch passage

Source `02-analysis.tex`, `lem:inverse`, uses $h = e^{-\beta s}$ with

$$3/2 < \beta < \min(\tau, \sqrt{3}), \qquad Lh = (\beta^2 - 2\beta N/r - 3)h \leq -(3 - \beta^2)h. \tag{2.1}$$

Its comparison estimate has constants independent of angular cutoffs. Finite angular type reduces the inverse to radial equations $y'' + 2(N/r)y' - (3 + \lambda/r^2)y = F_\lambda$; a forcing exponent greater than three yields $y = r^{-3}\zeta + o(r^{-3})$. This is the source of differentiated end data, not an inference from a weighted uniform remainder. The source does not claim all high radial derivatives of a general seed decay.

Source `lem:green` first takes the mass radius limit at a fixed actual branch member. The discarded nonlinear flux terms are $O(r^{3-2\tau}) + O(r^{-\tau})$, tending to zero. Only afterwards does the source expand convergent Green integrals. Source `lem:rotation` averages rotated TT tensors with nonnegative polynomial kernels on $SO(3)$; this preserves TT and any boundary sign and gives convergence in $\|T\|_\beta = \sup e^{\beta s}|T|$ using the strict margin $\beta < \tau$. No branch for an approximating seed is assumed. Source `prop:expansions` proves three separate actual remainders: the function in weight β, its interior equation in weight 2β, and its boundary equation in the uniform norm. All are $O(\epsilon^{p+1})$, with $p = 2$ for all seeds and $p = 4$ on the specific kernel. These distinctions are essential.

2.2 CMC jets, quadratic cancellation, and exact kernel

Source `03-cmc.tex`, `prop:cmc-jets`, constructs only finite jets of CMC spheres. At finite radius their nonconstant Jacobi multipliers are

$$J_s^0 = -\Delta_{r^2\sigma} - 2/r^2 + 6m_0/r^3, \qquad \lambda_\ell(s) = (\ell(\ell+1) - 2)/r^2 + 6m_0/r^3. \tag{2.2}$$

All are positive for $\ell \geq 1$, but the dipole multiplier tends to zero at infinity. Rescaled boost variables are therefore part of the proof, not an optional numerical conditioning change. The source identifies the flux aspect $(C + 3E)/2$, transforms it under Lorentz boosts, and selects the frame with zero spatial mass. This is what makes the CMC Hawking mass limit the Lorentz norm, not merely p_0. The derivative budget at order p is $p - j + 1$ derivatives of metric coefficient j.

Source `lem:hawking-variation` gives

$$m'_H = \sqrt{A/(16\pi)}\frac{HA}{8\pi}\left[\bar{f}\left\langle \frac{R_G + 6 + |\mathrm{II}^\circ|^2}{2}\right\rangle + \bar{f}^{-1}\langle \delta f J \delta f\rangle\right]. \tag{2.3}$$

Here brackets are the current leaf area average, f is the normal lapse, and $\delta f = f - \bar{f}$. The source integrates each coefficient first on $[0, T]$ and then takes the endpoint limit. Nonnegativity of the resulting sphere-integrated terms proves convergence; it is not obtained by an unjustified exchange of differentiation with an improper integral.

Let $Q(v) = \operatorname{Hess} v - Bv$, $Lv = 0$, $v|_S = D^{-1}q_{ss}$. Source `04-quadratic.tex`, `lem:tt-square`, proves

$$\int_M N|q|^2 \mathrm{d}V_0 - \kappa \int_S vDv \mathrm{d}A_0 = \int_M N|q - Q(v)|^2 \mathrm{d}V_0. \tag{2.4}$$

The minimal-leaf area differs from the original boundary area by $\epsilon^2 \int_S vDv/2$ at degree two. Combining this with $b'_*(A_0)$ and (2.3) gives, for finite angular type,

$$16\pi c_2[q] = \int_M N\left(|q - Q(v)|^2 + |U_1|^2 + 2F_1 J_s^0 F_1\right)\mathrm{d}V_0. \tag{2.5}$$

The F term is positive after sphere integration, not pointwise. Rotation approximation extends nonnegativity of c_2; it does not assert pointwise convergence of CMC lapse or shear fields for arbitrary seeds.

The source's exact kernel is

$$\mathcal{K} = \{Q(v) : Lv = 0,\ v \to 0,\ v|_S = c + d \cdot x,\ c \in \mathbb{R},\ d \in \mathbb{R}^3\}. \tag{2.6}$$

Necessity uses the vanishing tracefree boundary Hessian, which excludes $\ell \geq 2$. Sufficiency also needs the source's dipole cancellation: $h_1 = -2r^3(u_1' - (N/r)u_1)/(3m_0)$ satisfies $h_1' = -2u_1$, so the first lapse fluctuation vanishes. Closedness and rotation invariance of the quadratic nullspace exclude additional arbitrary-angular null modes. The sign condition $q_{ss}|_S \geq 0$ would read $D_0 c \geq D_1|d|$, $D_1 = 6m_0/a^3$, and does not exclude mixed dipoles.

2.3 Fourth order and radial equality

Source `05-quartic.tex` works in the algebra of coefficients modulo ϵ^5. Its Gaussian time recursion is

$$\partial_t \gamma = 2\mathcal{K}_t, \qquad \partial_t \mathcal{K}_t = -\operatorname{Ric}_\gamma - 3\gamma - (\operatorname{tr}_\gamma \mathcal{K}_t)\mathcal{K}_t + 2\mathcal{K}_t \circ_\gamma \mathcal{K}_t. \tag{2.7}$$

The contracted Bianchi recursion propagates the constraint residuals as a formal time series. A full formal time series is retained before substituting $t = T = -\epsilon v + \epsilon^2 w$; differentiating an already discarded time term as zero would be invalid. With $u = u_1$,

$$Lw = 2\langle \mathrm{d}u, \mathrm{d}v\rangle - 12uv, \quad w|_S = 0, \tag{2.8}$$

$$k = Q(w) - 2uq - 2v \operatorname{Hess} u + 6uv g_0 + 2\mathrm{d}u \otimes \mathrm{d}v + 2\mathrm{d}v \otimes \mathrm{d}u - 2\langle \mathrm{d}u, \mathrm{d}v\rangle g_0. \tag{2.9}$$

The source derives $\widetilde{K} = \epsilon^2 k + O(\epsilon^3)$ with k TT, $R_{\widetilde{g}} + 6 = \epsilon^4|k|^2 + O(\epsilon^5)$, and $\widetilde{g} - g_\epsilon = O(r^{-6})$ through the required degrees. Its explicit second metric change is

$$[\widetilde{g} - g_\epsilon]_2 = -\operatorname{Hess}(v^2) + \mathrm{d}v \otimes \mathrm{d}v + Bv^2. \tag{2.10}$$

The signs and derivative terms in this formula cannot be inferred from scalar jet closure alone.

The null-cut lemma starts with $\chi(0,\epsilon) = O(\epsilon^2)$ and zero background null screen curvature. It gets $\chi = O_{\mathrm{tot}}(2)$, $\theta = O_{\mathrm{tot}}(5)$; a cut at affine parameter $O(\epsilon)$ consequently preserves boundary area through degree four. Variable cut parameters do not add screen metric terms because the null generator is orthogonal to the transported angular fields. One-sided boundary jets justify formally negative displacements; no actual constrained extension through S is supplied. The subsequent minimal-height difference starts at $\epsilon^2 D^{-1} k_{ss}$. The source concludes, with $Lv_k = 0$ and $v_k|_S = D^{-1}k_{ss}$,

$$c_3 = 0, \qquad 16\pi c_4 = \int_M N\left(|k - Q(v_k)|^2 + |U_2|^2 + 2F_2 J_s^0 F_2\right) \mathrm{d}V_0. \tag{2.11}$$

This requires all the preceding transformed-slice and null-cut identities.

If this quartic expression vanishes, source `lem:round-jet` rounds the constant-curvature angular two-jet and produces local conformal tangency $\mathrm{d}v \otimes \mathrm{d}v + Bv^2 = \mathcal{L}_Y g_0 + c g_0$ on each compact interior interval. The degree-two projection of a dipole then implies

$$(V' - (N/r)V)^2 = (6m_0/r^3)V^2. \tag{2.12}$$

Decay excludes a nonzero V. Section 6 proves a residual version of this specific projection and final ODE argument; it does not independently prove that equality in the geometric quartic expression gives the required conformal tangency.

Source `06-conclusion.tex` treats radial seeds exactly: $q = C_q r^{-3}(\mathrm{d}s^2 - r^2\sigma/2)$. Small-solution uniqueness against every rotation first establishes radiality of the actual branch. With $\mathrm{d}\ell = \phi^2 \mathrm{d}s$, $R = r\phi^2$, and $K = -2k_r \mathrm{d}\ell^2 + k_r R^2 \sigma$, $k_r = -\epsilon C_q/(2R^3)$, the conserved quantity is

$$\mathcal{M} = \frac{R}{2}(1 + R^2 - \dot{R}^2 + R^2 k_r^2). \tag{2.13}$$

The scalar and momentum constraints give $\dot{\mathcal{M}} = 0$ without dividing by $\dot{R}$. The boundary relation $\dot{R}/R + k_r = 0$ gives $\mathcal{M} = b_*(A)$, and $\phi = 1 + cr^{-3} + o(r^{-3})$ gives the same flux mass $m_0 + 8c$. This is an exact radial conservation argument, not a consequence of any finite number of vanishing deficit coefficients.

3 Exact inexact-potential TT identity

Here is a local extension proved independently from the static equations. Write $\|T\|_N^2 = \int_M N|T|^2 \mathrm{d}V_0$ and let boundary pairings use $\mathrm{d}A_0$. All statements in this section are real-valued.

Theorem 3.1 (Residual-exact square)**.** *Let t be a smooth symmetric tensor with* $\operatorname{div} t = 0$ *and $|t| = O(r^{-\gamma})$, $\gamma > 3/2$. Let w be smooth up to S, with covariant derivatives through order three $O(r^{-\delta})$, $\delta > 3/2$. Put $h = t_{ss}|_S$, $f = Lw$, and $e = D(w|_S) - h$. Then*

$$\boxed{\|t\|_N^2 - \kappa\left\langle h, D^{-1}h\right\rangle_S = \|t - Q(w)\|_N^2 - \|Lw\|_N^2 - \kappa\left\langle e, D^{-1}e\right\rangle_S.} \tag{3.1}$$

No tracefree assumption is required for this identity. If additionally t is TT and v is its exact decaying harmonic lifting, the left side equals $\|t - Q(v)\|_N^2$. Thus the approximate potential square always overestimates *this exact TT square by the two displayed nonnegative residual terms.*

Proof. For an arbitrary w, the static equations give

$$\operatorname{tr} Q(w) = f, \quad \operatorname{div} Q(w) = \mathrm{d}f, \quad \omega = N\mathrm{d}w - w\mathrm{d}N, \quad \operatorname{sym}\nabla\omega = NQ(w), \quad \operatorname{div}\omega = Nf. \tag{3.2}$$

The symmetric gradient is averaged. The derivative commutator for a function yields $\operatorname{div}\operatorname{Hess} w = \mathrm{d}\Delta w + \operatorname{Ric}(\nabla w, \cdot)$, so the first two identities follow directly; mixed gradient terms cancel in the third. At the totally geodesic boundary, $Q(w)_{ss} = Dw + f$ and $\omega(-n) = \kappa w$. Integration by parts first with t, then with $Q(w)$, gives

$$\langle t, Q(w)\rangle_N = \kappa \langle w, h\rangle_S , \tag{3.3}$$

$$\|Q(w)\|_N^2 = \kappa \langle w, Dw + f\rangle_S - \int_M \langle \mathrm{d}f, \omega\rangle \, \mathrm{d}V_0 = \kappa \langle w, Dw\rangle_S + \|f\|_N^2 . \tag{3.4}$$

For the last equality integrate the $\mathrm{d}f$ term once more using $\operatorname{div}\omega = Nf$; its inner boundary term is $\kappa \langle f, w\rangle_S$, which cancels exactly. The outer boundary terms have orders $r^{3-\gamma-\delta}$ and $r^{3-2\delta}$ and vanish. The same exponents give absolute convergence of the relevant bulk pairings. Expanding $\|t - Q(w)\|_N^2$ now yields the left side of (3.1) plus $\|f\|_N^2$ and

$$\kappa \left\langle w - D^{-1}h, D(w - D^{-1}h)\right\rangle_S = \kappa \left\langle e, D^{-1}e\right\rangle_S .$$

Positivity follows from the spectrum of D. For the harmonic lifting, (3.2) makes $Q(v)$ TT and $Q(v)_{ss} = h$; (2.4) also follows by setting $w = v$ in the identity just proved. For finite angular data the lifting exists by the separated Dirichlet construction described above; the assertion about a lifting is conditional on its existence in any broader class. □

Corollary 3.2 (Usable lower certificate)**.** *For an exact finite-angular TT seed q in the source quadratic identity, suppose rigorous computations enclose $\|q - Q(w)\|_N^2 \geq Z_-$, $\|Lw\|_N^2 \leq R_+$, and $\langle e, D^{-1}e\rangle_S \leq E_+$. Then*

$$c_2[q] \geq (Z_- - R_+ - \kappa E_+)/(16\pi). \tag{3.5}$$

The same lower bound with q replaced by the exact transformed TT tensor k applies to c_4 on the exact quadratic kernel, conditional on *the source identity* (2.11).

Proof. Substitute (3.1) in the relevant identity and drop its nonnegative shear and sphere-integrated lapse terms. □

This bound deliberately need not detect every positive coefficient: the omitted geometric terms can carry all its positivity. It replaces a generic Hessian error bound with the exact scalar residual norm, and it uses no coercivity constant for recovering $\operatorname{Hess}(w - v)$ from Lw. It also avoids dividing by N, which vanishes at the horizon. With $L^2(S)$-orthonormal spherical harmonics, $\langle e, D^{-1}e\rangle = \sum_{\ell,m} |e_{\ell m}|^2/D_\ell$; in particular it is at most $\|e\|^2_{L^2(S)}/D_0$. Unresolved high harmonics cannot simply be set to zero.

3.1 Constraint defects and finite Gram matrices

If t is not divergence-free but the integrations above remain justified, the right side of (3.1) acquires

$$-2\int_M \langle \operatorname{div} t, N\mathrm{d}w - w\mathrm{d}N\rangle \, \mathrm{d}V_0. \tag{3.6}$$

This follows by retaining the omitted integration-by-parts term in (3.3). It is bounded by twice the integral of the product of the two absolute values, whenever that majorant is integrable. A weighted N^{-1} Cauchy–Schwarz estimate is not automatically finite at S. Trace and divergence defects also affect whether t represents an admissible physical seed at all. Equation (3.6) does not extend the source Penrose theorem to non-TT seeds without a proved TT correction and an error bound for changing the seed.

For a finite basis of residual tensor fields Z_j, the physical matrix is $G_{ij} = \int_M N \langle Z_i, Z_j\rangle \, \mathrm{d}V_0$, not the Euclidean identity. For the combined fields in (2.5) one adds the shear matrix and

$2\int NF_iJ_s^0F_j$ with sphere integration first. This Gram matrix is positive semidefinite; it is positive definite only after linear dependence and the relevant null directions have been removed. If $G_0 > 0$ is a declared reference metric and a verified approximation obeys $|a^T(G-\widehat{G})a| \le \eta a^T G_0 a$, then $a^T G a \ge a^T\widehat{G}a - \eta a^T G_0 a$ for every coefficient vector. This one-line estimate charges quadrature, rounding, and tail errors only when they have actually been bounded. A finite radial or harmonic cutoff does not prove that the infinite-dimensional omitted component is small.

4 Explicit residual-to-branch and mass bounds

We now quantify the analytic step between an approximate jet and a given actual solution. This is an a posteriori comparison, not an existence theorem. Put $\psi = \phi_\epsilon - 1$, $h_q = q_{ss}|_S$, $Q_q = |q|^2$, and

$$\begin{aligned}\mathcal{A}(z) &= \tfrac{15}{2}z^2 + \tfrac{15}{2}z^3 + \tfrac{15}{4}z^4 + \tfrac{3}{4}z^5,\\ \mathcal{N}(z) &= \mathcal{A}(z) - \tfrac{\epsilon^2}{8}Q_q(1+z)^{-7}, \qquad \mathcal{B}(z) = \tfrac{\epsilon}{4}h_q(1+z)^{-3}.\end{aligned} \tag{4.1}$$

Choose a smooth trial P, such as $\sum_{j=1}^p \epsilon^j u_j$, and define

$$R_P = LP - \mathcal{N}(P), \qquad b_P = P_s|_S - \mathcal{B}(P|_S). \tag{4.2}$$

Both P and ψ must be $o(e^{-\beta s})$, and we assume a known tube $|P|, |\psi| \le d < 1$ and $|P|, |\psi| \le Me^{-\beta s}$. The source convergence guarantees some small tube eventually but gives no numerical radius for a particular seed. Define

$$\begin{aligned}H(d) &= 15 + \tfrac{45}{2}d + 15d^2 + \tfrac{15}{4}d^3,\\ v_* &= H(d)d + \tfrac{7\epsilon^2}{8}\|q\|_\infty^2 (1-d)^{-8},\\ a_* &= \tfrac{3|\epsilon|}{4}\|h_q\|_\infty (1-d)^{-4}, \quad K_* = H(d)M + \tfrac{7\epsilon^2}{8}\|q\|_\beta^2 (1-d)^{-8}.\end{aligned} \tag{4.3}$$

Theorem 4.1 (Explicit branch comparison)**.** *Under the preceding hypotheses, if* $\lambda = 3 - \beta^2 - v_* > 0$ *and* $\mu = \beta - a_* > 0$*, then*

$$\|\psi - P\|_\beta \le E_P := \max\left\{\frac{\|R_P\|_\beta}{\lambda}, \frac{\|b_P\|_\infty}{\mu}\right\}. \tag{4.4}$$

Moreover

$$\|\mathcal{N}(\psi) - \mathcal{N}(P)\|_{2\beta} \le K_* E_P, \qquad \|\mathcal{B}(\psi) - \mathcal{B}(P)\|_\infty \le a_* E_P. \tag{4.5}$$

Proof. For $e = \psi - P$ the integral mean value formula gives $(L-V)e = -R_P$, $e_s - ae = -b_P$, where V and a are the derivatives of $\mathcal{N}$ and $\mathcal{B}$ averaged on the segment between P and ψ. The segment stays in the declared tube. From $|\mathcal{A}'(z)| \le H(d)|z|$ we obtain $|V| \le v_*$, $|a| \le a_*$. Divide by $h = e^{-\beta s}$. The resulting interior operator has zeroth-order coefficient $\beta^2 - 2\beta N/r - 3 - V \le -\lambda$. At a positive interior maximum of e/h exceeding $\|R_P\|_\beta/\lambda$ its equation is impossible. At S the boundary condition reads $(e/h)_s = (\beta+a)e/h - b_P$; a boundary maximum larger than $\|b_P\|_\infty/\mu$ would have positive inward derivative, also impossible. The ratio tends to zero at infinity, so positive maxima can be localized to a compact set. Apply the same argument to $-e$ and, if necessary, use a strict bound then take its limit. This proves (4.4). The sharper pointwise bound $|V(s,x)| \le K_* e^{-\beta s}$ follows from the second tube estimate and $|q|^2 \le \|q\|_\beta^2 e^{-2\beta s}$. Multiplication by $|e|$ gives the first estimate in (4.5); the boundary derivative bound gives the second. □

4.1 The convergent observable and its error

Define $W_0 = N$, $W_i = rx_i$, so $LW_0 = 0$ and $LW_i = -3m_0 r^{-2} x_i$. For triples with the stated decay put

$$\mathcal{G}_\mu(u, F, b) = \frac{1}{2\pi}\left[\int_S (uW_{\mu,s} - W_\mu b)\mathrm{d}A_0 - \int_M (W_\mu F - uLW_\mu)\mathrm{d}V_0\right]. \tag{4.6}$$

The Green divergence identity and the fixed-member flux estimates in Section 2 give exactly $p_\mu - p_\mu(0) = \mathcal{G}_\mu(\psi, \mathcal{N}(\psi), \mathcal{B}(\psi))$. There is no derivative of the unknown branch remainder on the right. To make its constants fully specified, set

$$\begin{aligned} C_{\mu u} &= \tfrac{1}{2\pi}\left(\int_S |W_{\mu,s}|\mathrm{d}A_0 + \int_M e^{-\beta s}|LW_\mu|\mathrm{d}V_0\right), \\ C_{\mu F} &= \tfrac{1}{2\pi}\int_M |W_\mu| e^{-2\beta s}\mathrm{d}V_0, \qquad C_{\mu b} = \tfrac{1}{2\pi}\int_S |W_\mu|\mathrm{d}A_0. \end{aligned} \tag{4.7}$$

All are finite since $2\beta > 3$. These exact one-dimensional/sphere integrals are definitions of constants, not claimed evaluated numbers. For the equation-consistent approximation

$$\widehat{p}_\mu = p_\mu(0) + \mathcal{G}_\mu(P, \mathcal{N}(P), \mathcal{B}(P)), \quad \widehat{A} = \int_S (1+P)^4 \mathrm{d}A_0, \tag{4.8}$$

Theorem 4.1 implies

$$|p_\mu - \widehat{p}_\mu| \le (C_{\mu u} + C_{\mu F}K_* + C_{\mu b}a_*)E_P =: \eta_\mu, \quad |A - \widehat{A}| \le 4A_0(1+d)^3 E_P =: \eta_A. \tag{4.9}$$

The value $\widehat{p}$ is an observable approximation. Unless the residuals vanish, it is not automatically the metric mass of $(1+P)^4 g_0$; the latter would use LP, P_s rather than $\mathcal{N}(P), \mathcal{B}(P)$ in the Green formula.

For an explicitly convex future-timelike tube $p_0 \in [p_-, p_+]$, $|p_{\mathrm{sp}}| \le s_+ < p_-$ containing p and $\widehat{p}$, put

$$L_m = \frac{\sqrt{p_+^2 + s_+^2}}{\sqrt{p_-^2 - s_+^2}}, \qquad L_A = \max_{A \in [A_-, A_+]} \frac{1 + 3A/(4\pi)}{8\sqrt{\pi A}},$$

where $0 < A_- \le A, \widehat{A} \le A_+$. The Euclidean norm of the gradient of $\sqrt{p_0^2 - |p_{\mathrm{sp}}|^2}$ is at most L_m there. The mean value theorem therefore proves

$$\left|\mathfrak{D}_q - \left(\sqrt{\widehat{p}_0^2 - |\widehat{p}_{\mathrm{sp}}|^2} - b_*(\widehat{A})\right)\right| \le L_m \sqrt{\sum_{\mu=0}^{3} \eta_\mu^2} + L_A \eta_A. \tag{4.10}$$

Any verified quadrature errors in (4.8) must be added to the component error bounds. A mass-component enclosure reaching the light cone cannot use this Lipschitz constant. Equations (4.4)–(4.10) are finite-parameter certificates when all tube, residual, tail, and quadrature inputs are known; this report does not claim they have been instantiated for a numerical seed.

4.2 Residual size, explicit radii, and end tails

If $\|R_P\|_{2\beta} \le C_R \epsilon^{p+1}$ and $\|b_P\| \le C_b \epsilon^{p+1}$ throughout an admitted interval with $\lambda \ge \lambda_0 > 0$, $\mu \ge \mu_0 > 0$ and the other constants bounded, then all errors above are $O(\epsilon^{p+1})$ with a displayed computable constant. This is the quantified version of the source's branch passage. It does not follow from a formal residual with no norm bound. If a coefficient computation and Taylor remainder bound yield

$$|\mathfrak{D}_q - c_2\epsilon^2| \le C_3\epsilon^3,$$

then a verified $c_2 \geq b_2 > 0$ gives $\mathfrak{D}_q \geq b_2\epsilon^2/2$ for $0 < \epsilon \leq b_2/(2C_3)$ within that interval. On the exact kernel, with $c_3 = 0$, the analogous conditions $c_4 \geq b_4 > 0$, $|\mathfrak{D}_q - c_4\epsilon^4| \leq C_5\epsilon^5$ give $\epsilon \leq b_4/(2C_5)$. If the remainder constant is zero the associated restriction is unnecessary. These are conditional radii, not measured ones.

An elementary explicit tail bound is available without solving for $s(r)$: $r' \leq \sqrt{1+r^2}$ implies $r(s) \leq \sinh(s + \operatorname{arsinh} a) \leq C_r e^s$ with $C_r = e^{\operatorname{arsinh} a}/2$, and $N \leq C_N e^s$ with $C_N = \sqrt{1+C_r^2}$. Thus for $|T| \leq He^{-\beta s}$,

$$\int_{s\geq S} N|T|^2 \mathrm{d}V_0 \leq \frac{4\pi C_N C_r^2 H^2}{2\beta - 3} e^{-(2\beta-3)S}. \tag{4.11}$$

The same estimate bounds time-test Green tails, while spatial-test and LW_i tails have the stated additional decay. This is a rigorous truncation formula, not evidence that a finite quadrature already captures the full exterior. The deterioration as $\beta \downarrow 3/2$ is real.

5 Complete coefficient recursion and its checks

Let $P_p = \sum_{j=1}^{p} \epsilon^j u_j$, with coefficients not divided by an extra factorial, and solve $Lu_j = F_j$, $(u_j)_s|_S = B_j$, $u_j \to 0$. The recurrence is triangular in parameter degree, not in radial polynomial degree. Direct expansion of (4.1) gives

$$\begin{aligned}
&F_1 = 0, \qquad F_2 = \tfrac{15}{2}u_1^2 - Q_q/8,\\
&F_3 = 15u_1u_2 + \tfrac{15}{2}u_1^3 + \tfrac{7}{8}Q_q u_1,\\
&F_4 = 15u_1u_3 + \tfrac{15}{2}u_2^2 + \tfrac{45}{2}u_1^2u_2 + \tfrac{15}{4}u_1^4 + \tfrac{7}{8}Q_q u_2 - \tfrac{7}{2}Q_q u_1^2,\\
&B_1 = h_q/4, \qquad B_2 = -3h_q u_1/4,\\
&B_3 = \tfrac{h_q}{4}(-3u_2 + 6u_1^2), \qquad B_4 = \tfrac{h_q}{4}(-3u_3 + 12u_1u_2 - 10u_1^3).
\end{aligned} \tag{5.1}$$

All boundary products are restricted to S. The fourth-order construction here is used only in the source's finite-angular, fully differentiated kernel class. We do not assume general high-order spatial decay for arbitrary q.

For reproducible coefficient extraction of the mass norm write $p_0 = m_0 + \sum a_j\epsilon^j$, $p_{\mathrm{sp}} = \sum b_j\epsilon^j$ and $m_{\mathrm{AH}} = m_0 + \sum M_j\epsilon^j$. Squaring gives

$$M_j = a_j + \frac{1}{2m_0}\sum_{i=1}^{j-1}(a_i a_{j-i} - b_i \cdot b_{j-i} - M_i M_{j-i}). \tag{5.2}$$

In particular $M_2 = a_2 - |b_1|^2/(2m_0)$; ignoring spatial mass would give the wrong quadratic coefficient. The area coefficients through four are integrals of

$$\begin{aligned}
&4u_1, \qquad 4u_2 + 6u_1^2,\\
&4u_3 + 12u_1u_2 + 4u_1^3,\\
&4u_4 + 12u_1u_3 + 6u_2^2 + 12u_1^2u_2 + u_1^4.
\end{aligned}$$

For $A = A_0 + \sum A_j\epsilon^j$, the coefficient of degree four in $b_*(A)$ is

$$b_*'A_4 + b_*''(A_1A_3 + A_2^2/2) + \tfrac{1}{2}b_*'''A_1^2A_2 + \tfrac{1}{24}b_*''''A_1^4, \tag{5.3}$$

all derivatives evaluated at A_0. These formulas provide a second scalar route for checking geometric coefficient identities, but the radial ODE solutions and infinite-volume integrals still need certification.

Parameter differentiation and spatial differentiation must not be confused. For the radial operator on harmonic degree ℓ,

$$L_\ell = \partial_s^2 + 2(N/r)\partial_s - 3 - \ell(\ell+1)/r^2,$$

every differentiated row contains derivatives of N/r and r^{-2}. The private full-Leibniz jet rule justifies retaining these terms; it gives no finite nilpotent inverse since these coefficient functions are not polynomials satisfying its degree-lowering hypotheses. Finite angular type is not finite radial dimension. The exterior boundary condition at infinity also cannot be recovered from one local derivative jet.

6 Residual form of the dipole obstruction

Fix $v = v_0(s) + V(s)x$, $Y_2 = x^2 - 1/3$, and set $P_0 = N/r$. For the electric degree-two projection of $\mathrm{d}v \otimes \mathrm{d}v + Bv^2 - (\mathcal{L}_Y g_0 + cg_0)$, let the four scalar residuals be

$$\begin{aligned} E_1 &= V'^2 + B_s V^2 - 2p' - c, \\ E_2 &= VV'/2 - p - r^2 d', \\ E_3 &= V^2/2 - 2r^2 d, \\ E_4 &= V^2 + r^2 B_t V^2 - 2rNp - r^2 c. \end{aligned} \tag{6.1}$$

Here $pY_2\partial_s + d\nabla_\sigma Y_2$ is the tested vector-field part. Products with v_0 have only degrees zero or one. The identity $\mathrm{d}x \otimes \mathrm{d}x = \mathrm{Hess}_\sigma Y_2/2 + x^2\sigma$ explains the Hessian and scalar entries without suppressing angular derivatives.

Proposition 6.1 (Residual elimination)**.** *For smooth profiles as above, put $P_E = E_3'/2 - (N/r)E_3 - E_2$. Then the exact identity is*

$$(V' - P_0 V)^2 - \frac{6m_0}{r^3}V^2 = E_1 + 2P_E' - 2P_0 P_E - E_4/r^2 =: \mathcal{E}. \tag{6.2}$$

Consequently a small bound on the four undifferentiated residuals is insufficient: this certificate requires two radial derivatives of E_3 and one of E_2.

Proof. The last three equations of (6.1) solve to

$$d = \frac{V^2}{4r^2} - \frac{E_3}{2r^2}, \quad p = \frac{NV^2}{2r} + P_E, \quad c = \frac{3m_0 V^2}{r^3} - 2P_0 P_E - E_4/r^2.$$

Substitute into the first. With $N' = r + m_0/r^2$ and $N^2 = 1 + r^2 - 2m_0/r$, the residual-free part is precisely $(V' - P_0 V)^2 - 6m_0 V^2/r^3$. The remaining terms give (6.2). This uses no differential equation for V and is thus a local tensor-projection identity beyond the exact kernel. □

Theorem 6.2 (Quantitative decaying obstruction)**.** *Let $V/r \to 0$ and let $\mathcal{E}$ be as in (6.2). For any $s > 0$, assume $\int_s^\infty \sqrt{|\mathcal{E}(t)|}/r(t)\mathrm{d}t < \infty$. Then*

$$\frac{|V(s)|}{r(s)} \le \int_s^\infty \exp\left(\int_s^t \sqrt{6m_0}\, r(u)^{-3/2}\mathrm{d}u\right) \frac{\sqrt{|\mathcal{E}(t)|}}{r(t)}\mathrm{d}t. \tag{6.3}$$

In particular zero projected residuals on the end force $V = 0$ there; if V also solves the source homogeneous dipole equation, ODE uniqueness then gives $V \equiv 0$.

Proof. For $z = V/r$, (6.2) becomes $z'^2 = (6m_0/r^3)z^2 + \mathcal{E}/r^2$. Hence $|z'| \le a(s)|z| + b(s)$ with $a = \sqrt{6m_0}r^{-3/2}$, $b = \sqrt{|\mathcal{E}|}/r$. Backward Gronwall on $[s,T]$ gives $|z(s)| \le |z(T)|e^{\int_s^T a} + \int_s^T e^{\int_s^t a} b(t)\mathrm{d}t$. The exponential stays bounded because $r \asymp e^s$. Taking $T \to \infty$ proves the assertion. Absolute values make the argument valid across zeros of z without choosing a branch of a square root. □

Corollary 6.3 (Explicit end obstruction budget)**.** *Choose $s > 0$ with $R = r(s) \geq \max(1, 2m_0)$. Suppose $|\mathcal{E}(t)| \leq Er(t)^{-\nu}$ for every $t \geq s$, where $\nu > -2$, and $V/r \to 0$. Then*

$$E \geq (1 + \nu/2)^2 \exp\left(-\tfrac{4}{3}\sqrt{6m_0}R^{-3/2}\right) R^{\nu}|V(s)|^2. \tag{6.4}$$

For the source's differentiated $O(r^{-3})$ profile, $\nu = 6$ is a natural finite weighted residual class, and the prefactor is 16 times the displayed exponential. The profile normalization and global envelope must be supplied; finitely many samples cannot establish the latter.

Proof. For $r \geq R$, $N^2 = r^2 + 1 - 2m_0/r \geq r^2$, so $r(t) \geq Re^{t-s}$ and $\int_s^\infty \sqrt{6m_0}r^{-3/2} \leq \frac{2}{3}\sqrt{6m_0}R^{-3/2}$. The integral in (6.3) is at most

$$\frac{\sqrt{E}}{1 + \nu/2} \exp\left(\tfrac{2}{3}\sqrt{6m_0}R^{-3/2}\right) R^{-1-\nu/2}.$$

Squaring its bound for $|V(s)|/R$ and rearranging proves (6.4). The finite-endpoint version, before discarding the terminal value, is

$$\frac{|V(s)|}{r(s)} \leq e^{\int_s^T a} \frac{|V(T)|}{r(T)} + \int_s^T e^{\int_s^t a} \frac{\sqrt{|\mathcal{E}(t)|}}{r(t)} \mathrm{d}t, \qquad a(t) = \sqrt{6m_0}r(t)^{-3/2}.$$

Thus a validated terminal enclosure can be used in place of an exact end condition, with its term explicitly retained. □

For a reconstructed profile $\widehat{V}$ with certified $|V - \widehat{V}| \leq \delta_0$, $|V' - \widehat{V}'| \leq \delta_1$, set $w = \widehat{V}' - (N/r)\widehat{V}$ and $\delta_w = \delta_1 + (N/r)\delta_0$. Difference of squares gives the useful pointwise enclosure

$$|\mathcal{E}_V - \mathcal{E}_{\widehat{V}}| \leq 2|w|\delta_w + \delta_w^2 + \frac{6m_0}{r^3}(2|\widehat{V}|\delta_0 + \delta_0^2), \tag{6.5}$$

where $\mathcal{E}_V = (V' - (N/r)V)^2 - 6m_0V^2/r^3$. This follows by bounding each difference $a^2 - b^2$ as $2|b||a - b| + |a - b|^2$. A value-only error bound does not supply δ_1. The explicit end bound and profile enclosure were proposed in the parent's local note and independently rederived here; the overlapping Gronwall estimate was already present in the author draft.

This estimate quantifies the final obstruction once a conformal tangency residual and its differentiated bounds have been supplied. It is not a bound of $|d|$ directly in terms of c_4: the intervening geometric stability from small shear/lapse energy to a small tangency residual is not proved here, and would require additional estimates on the noncompact end. Its square-root dependence also warns against claiming a linear inverse near a quadratic obstruction.

7 Quartic symmetry and the near-kernel boundary

The following exact structural consequence is conditional on the source's coefficient construction and its sign/rigidity conclusions on $\mathcal{K}$. It does not presuppose a numerical value for a quartic constant.

Proposition 7.1 (Invariant quartic normal form)**.** *Use the boundary coordinates $(c, d) \in \mathbb{R} \times \mathbb{R}^3$ of (2.6). If the coefficient construction through order four is valid, c_4 is a homogeneous polynomial of degree four in (c, d) and is rotation invariant. If also the source's radial equality and strict positivity for $d \neq 0$ hold, there exist constants $A \geq 0$, $B_4 > 0$, depending on m_0 and this boundary normalization, such that*

$$c_4(c, d) = Ac^2|d|^2 + B_4|d|^4 \geq B_4|d|^4. \tag{7.1}$$

Proof. The linear inverse is fixed. Recursion (5.1) makes each u_j homogeneous polynomial of degree j in the seed coordinates; all products, Green functionals and scalar composition coefficients preserve the total degree. Thus c_4 is a degree-four polynomial. Rotation invariance follows from the round background and the Lorentz norm. A polynomial in one vector d invariant under rotations is a polynomial in $|d|^2$: on the axis it is even (a rotation sends the axis to its negative), and each sphere is a rotation orbit. Homogeneity therefore leaves only $Cc^4 + Ac^2|d|^2 + B_4|d|^4$. Radial equality forces $C = 0$. Strict positivity at $(0, d)$ gives $B_4 > 0$; nonnegativity for fixed $d \neq 0$ and arbitrarily large $|c|$ forces $A \geq 0$. □

The normalization matters: d is boundary Dirichlet amplitude, not a unit tensor coefficient in an unspecified basis. The result reduces a future invariant scalar check to two coefficients and provides a lower quartic bound independent of the radial component once B_4 is verified. It does not give a computed lower bound for B_4, nor a uniform branch radius near $d = 0$, nor an estimate off the exact kernel.

Indeed the elementary homogeneous model

$$F_{x,y}(\epsilon) = \epsilon^2 y^2 - 3\epsilon^3 x^2 y + \epsilon^4 x^4 \tag{7.2}$$

has nonnegative quadratic coefficient and strictly positive quartic coefficient on its nonzero null line $y = 0$. Every fixed (x, y) gives nonnegativity for sufficiently small positive ϵ, but $F_{1,\epsilon}(\epsilon) = -\epsilon^4 < 0$. It is a logical counterexample to uniform near-kernel inference, not a counterexample to the fixed-seed source theorem. It even respects degree-two, degree-three, and degree-four homogeneity in the seed coordinates. In a quantitative mixed estimate of the form $\epsilon^2\lambda t^2 - \epsilon^3 A_3 t + \epsilon^4 b_4$, completing the square requires $b_4 \geq A_3^2/(4\lambda)$ for uniform nonnegativity of this lower bound. Neither separate positivity of c_2 off its kernel nor c_4 on it proves that inequality. Near-kernel cubic terms must be bounded, not erased.

8 Specific private Ram correspondences

The following are actual passages read from local originals. They remain confidential and were not uploaded or quoted in public web queries. The mathematical applications below distinguish their representation rules from classical differential geometry and ordinary Taylor analysis.

Physical Gram metric and omitted residual. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch01d-physical-metrics.tex`, sections *A Nonorthogonal Representation Carries a Metric*, *The Full Residual Has an Omitted Component*, and *Approximating the Metric*; labels `v21c01:physical-gram-definition`, `v21c01:metric-symmetry`, `v21c01:residual-decomposition`, `v21c01:residual-identity`, `v21c01:gram-equivalence`. The entire file was read. Its synthesis J transports the norm as $G = J^*J$, and its full residual has an orthogonal omitted component. Here synthesis is into the weighted tensor, shear and mean-zero lapse fields of (2.5); the physical measure is $N\mathrm{d}V_0$ and the lapse block uses J_s^0. The exact new identity (3.1) is derived from the static equations, not quoted from that general Hilbert-space statement. The correspondence helps prevent treating a finite projected residual as a continuum norm.

Full differentiated jets and closure defects. `C:/Users/mural/OneDrive/Documents/RamNonlinearIE-PDE/TR1-lin-nonlin-rev2/sec-03-jets.tex`, subsections *Differentiated rows, moving-limit endpoint terms, and the differentiated remainder*, *Rank, degree closure, explicit matrices, and conditioning*, and *The nonlinear jet map, its Jacobian, and the closure operator*. The read passages include the complete statements and proofs of `thm:t1-leibniz-rows`,

`thm:t1-recovery`, `thm:t1-degree-closure`, `thm:t1-nonlinear-jetmap`, and the discussion through `eq:t1-jet-closure-defect`. They require all coefficient derivatives and retain the term discarded by finite closure. Here that requirement has a concrete manifestation in E_3'' and E_2' in (6.2) and in the radial derivatives of N/r in CMC and conformal jets. It is not evidence of finite polynomial closure for L.

Local moment remainder and polynomial exactness. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume1-rev2/chapters/ch03-local-moments-rev2.tex`, *Output-centered localization*, *Polynomial exactness is not polynomial closure*, and *Differentiating the local formula*; labels `thm:ram-local`, `eq:ram-integral-remainder`, `eq:ram-remainder-bound`, `thm:degree-closure`, `eq:ram-jet-product`. The selected theorem proofs and the complete closure/product-rule passages were read. Their integral remainder requires actual derivative and moment integrability bounds. In the present exterior PDE problem the appropriate replacement is an equation residual in weight 2β plus the boundary residual, followed by the explicit comparison and Green estimates. A noncompact Green kernel has not been converted here into a finite Ram moment inverse; no missing moment bounds are presumed.

Inexact identity certificates. `C:/Users/mural/OneDrive/Documents/MyDocuments2026/res-rt/res-math/TR-item-047-astra.tex`, *Global Optimality Sandwich*, theorem *Replayable global gap certificate*, equation `eq:gap`; and *Residual-Aware SOS Lower Bounds*, theorem *Inexact Gram and identity certificate*, equation `eq:gramerror047`. The proofs and the explicit residual example were read. The transferable principle is to subtract certified identity and matrix errors before declaring a positive lower bound. No SOS optimization problem has been substituted for the TT exterior problem. In particular, an interval containing zero does not certify the exact equality $c_2 = 0$, which is required before applying the source's special quartic branch.

These uses of Ram representations are compatible with classical Gram matrices, Taylor's theorem, integration by parts, maximum principles, and Gronwall inequalities. None of those classical techniques is attributed here as a historically new Ram discovery. The realized local contributions are the source-specific residual formulas and their proved consequences, not a claim of priority for the underlying methods.

9 Public comparison and research boundaries

Public primary literature was searched by public mathematical titles and topics on October 7, 2026; no private corpus text was sent to those services. Khuri–Kopiński [2] prove an applicable perturbative Penrose result under their Theorem 2 domination condition (2.5), comparing a lapse-weighted bulk seed norm with a boundary-neighborhood H^1 norm. Both sides have the same quadratic scaling. The pinned manuscript claims to remove that condition by exact TT cancellation and a separate quartic treatment; our residual formula quantifies the cancellation when its auxiliary potential is imperfect, but is not itself an independent proof of that removal.

Ambrozio [3] treats small Riemannian perturbations of positive-mass Schwarzschild–AdS using CMC geometry. The present source concerns nonzero extrinsic curvature and a MOTS boundary; replacing its argument with the time-symmetric theorem would not justify the boundary correction or quartic change of slice. Khuri–Kunduri [4] establish spacetime Penrose inequalities in specified cohomogeneity-one settings, including asymptotically hyperbolic analogues. Those symmetry hypotheses do not cover the arbitrary angular fixed seeds in M517. A current 2026 primary article on asymptotically locally hyperbolic static vacuum black holes [5] addresses static uniqueness and boundary geometry, not a substitute for the maximal

conformal-branch hypotheses here. This was a targeted comparison, not an exhaustive priority search or a claim that no other literature exists.

The new residual approach has three advantages: exact sign accounting for an inexact scalar lift, no weighted Hessian inverse constant in that square, and a direct route from actual equation residuals to the mass deficit. Its costs are equally specific: exterior tails must be bounded, the branch tube and timelike tube must be certified, and tensor constraints cannot be replaced by small unconstrained finite residuals. Scalar residual bounds alone do not validate the CMC boost construction, formal Einstein recursion, or quartic null transport.

The most useful next steps are a certified radial solver for one normalized dipole to bound B_4, a tensor-aware implementation of (6.1) retaining its differentiated residuals, and geometric stability estimates converting small quartic energy to approximate conformal tangency on an exterior domain. A uniform theorem near the kernel would additionally require mixed cubic estimates and uniform branch constants. None is reported as accomplished. No numerical speedup, test-seed success rate, or independently checked global theorem is claimed.

10 Verification record and contribution assessment

The source and private read scopes are recorded in `_audit/M517-author-inputs.json`; executed bounded symbolic checks and their limitations are recorded in `_audit/M517-author-checks.json`. The check script is owned by this report in its private build directory. The 30 author checks all passed under Python 3.9.13 and SymPy 1.12. They compare the third- and fourth-order conformal source and boundary coefficients, the mass-norm and area composition recurrences, static dipole elimination, its residual extension, and the radial conservation algebra. They are finite exact symbolic identities, not tests of PDE solutions or computer-assisted proof of the source's CMC/null-cut constructions.

The substantial source was made available before formatting and before the parent's final review. The independent audit at `_audit/M517-independent-audit.txt` reports 54 exact symbolic checks and no consequential detected error in its targeted source identities; its receipt distinguishes those tests from the general geometric arguments and records unperformed verification. That is independent supporting evidence, not an author claim to have executed those 54 tests or a formal verification of the whole source theorem. The author's separate executed tests are recorded in the author check file. The parent read the complete substantial report and found no consequential error in the residual square, branch comparison, Green readout, dipole elimination and stability, or the conditional quartic normal form within their declared assumptions. The parent note at `_audit/M517-parent-dipole.txt` supplied the explicit end constant and profile-error formulation integrated after independent derivation above.

The parent reported that native opening was queued and native compilation returned `compile-failed`, with the exact diagnostic

```
Unable to find standard directories for platform
```

followed by a newline. No native compiler success is claimed. The disk PDF was exported with the existing installed-MiKTeX helper and inspected on every page through contact sheets, with detailed samples of the principal identities and end estimates. The second formatting round corrected long paths, display widths, and contents pagination. The accompanying author audit records the exact export, page, warning and hash results. No software was installed. Public identifiers were checked against primary metadata, including the title, authors, journal reference and DOI of arXiv:2209.01234.

The realized contribution is assessed as C3/5 under the local policy: Theorem 3.1, Theorem 4.1, (4.10), Proposition 6.1, and Theorem 6.2 supply exact local improvements to the

verification mechanism. Proposition 7.1 is a useful conditional structural consequence, not independent quartic rigidity. The score is not the catalogue applicability level, not external peer review, and not credit for inherited source claims. The remaining geometric dependencies and absence of an instantiated numerical certificate prevent a claim of a fully verified nonlinear Penrose theorem.

References

[1] OpenAI, *A local Penrose inequality for conformal perturbations of Schwarzschild–anti-de Sitter data*, October 5, 2026, local source pinned at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

[2] M. Khuri and J. Kopiński, *Asymptotically hyperbolic Einstein constraint equations with apparent horizon boundary and the Penrose inequality for perturbations of Schwarzschild-AdS*, Classical and Quantum Gravity **40** (2023), 045007. arXiv:2209.01234; author-hosted full text, Theorem 2 and (2.5).

[3] L. C. Ambrozio, *On perturbations of the Schwarzschild anti-de Sitter spaces of positive mass*, Communications in Mathematical Physics **337** (2015), 767–783. arXiv:1402.4317.

[4] M. Khuri and H. Kunduri, *The spacetime Penrose inequality for cohomogeneity one initial data*, primary preprint (2024). arXiv:2404.13247.

[5] B. Harvie and Y.-K. Wang, *On the Geometry and Uniqueness of Asymptotically Locally Hyperbolic Static Vacuum Black Holes*, Communications in Mathematical Physics **407** (2026), article 198, published August 1. doi:10.1007/s00220-026-05717-y.